

Question number	Answer	Additional guidance	Mark
(a)	A description that includes two of the following points: • {hypertension / high blood pressure} damages (brain) {capillaries / endothelium / blood vessels} (1) • leading to blood vessel {narrowing / blockage / bleeding} (1) • reducing oxygen(ated blood) to brain (cells) / causes inflammation in the brain (1)	ignore microstructural damage without reference to blood vessel ACCEPT formation of {atheroma / plaques} / deposition of amyloid ACCEPT damages areas of the brain responsible for thinking and memory ACCEPT smaller brain volume	(2)

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(b)	An answer that includes the following points: • beta-amyloid is fat soluble (1) • {hydrophobic / hydrophilic / polar / non-polar / charged } (R groups of protein) (1) • (which cause) bonds to be formed between protein (molecules to clump them together) (1)	reject {hydrophobic / hydrophilic / polar / non-polar / charged} {tails / fatty acids} ACCEPT which affect the bonds made (with other proteins) ACCEPT named bonds but ignore incorrect bonds or bonds within a protein molecule ACCEPT hydrophobic regions (on different proteins) clump together (to move away from water) ACCEPT {hydrophobic / non-polar / hydrophilic / polar} interactions (with other proteins or water)	(3)

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(c)	An answer that includes the following points: • {2D / 3D / computer} image (1) • identify areas of brain activity / contrast between tissues / identify areas of {oxygen / glucose} uptake / identify plaques / identify amyloid build up (1) • provides detailed {structural / anatomical} (information) (1)	ignore high resolution image ACCEPT identify {biochemical changes / metabolic activity} in the brain	(2)

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(d)	An answer that includes one of the following points: • Suitable reason stated (1)	e.g. risk isn't certain / findings are unclear / could be false positive / they may not develop it after all e.g. may affect {health / mental health / behaviour} in the short term E.g. may affect getting health insurance (1) e.g. may affect {relationships / work / career opportunities} (1) e.g. may cause {stress/ fear / worry / anxiety / distress / depression} e.g. they have right to decide if they want to know	(1)

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(e)	An answer that includes three of the following points: • (low blood pressure) leads to reduced blood flow to the brain (1) • reduced {oxygen / glucose} (to the brain cells) (1) • leading to decrease in {aerobic respiration / ATP production} (1) • leading to reduced activity (in the brain) / {death of / damage to / loss of} brain cells (1)	ACCEPT reduced {amino acids / fatty acids / glycerol} to brain (cells) ignore nutrients / minerals ACCEPT reduced {protein / cell / myelin sheath} formation in brain ignore anaerobic respiration ACCEPT reduction in {neurotransmitter/ dopamine} secretion ignore reduction in brain volume ignore damage to endothelial cells	(3)

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(f)	A description that includes the following points: • {sequence the genomes / / use DNA / mRNA} of people with and without dementia (1) • suitable method (1) • {identify / analyse} {alleles / genes} only found in people with dementia (1) • identify people with family history of dementia (1)	e.g. use of FISH / use of microarrays / specific probes/hybridization / use of UV fluorescence to identify gene Also accept use of PCR {identify / analyse} active genes (in dementia)	(3)

Question number	Answer	Additional guidance	Mark
8(g)	A description that includes two of the following points: • an antibody that {binds to / complementary to} a (specific) antigen (1) • produced by a {clone of B lymphocyte / plasma cell / cell line} (1) • causes {agglutination / phagocytosis} (1)	ignore identify antigen ACCEPT forms antibody-antigen complex ACCEPT antibody used for opsonization ACCEPT antibody that is specific to an antigen ACCEPT produced by hybridoma / (description of) formation of hybridoma ACCEPT appropriate genetic engineering methods ignore produced by immune cell	(2)

Question number	Answer	Additional guidance	Mark
8(h)	An explanation that includes one of the following points: • {to act as a control / as a method of control} (1) • to compare with the group which received {the drug / donanemab} (1) • remove {bias / psychological effect} (1) • without a placebo group to compare against, it is not possible to know whether the treatment itself had any effect (1)	ACCEPT make results statistically valid / comparable ACCEPT to compare the {results / effectiveness} / for comparison / provide contrast to drug ACCEPT to prove it was {the drug / donanemab} that had the effect ACCEPT show effect of {drug / donanemab}	(1)

Question number	Answer	Additional guidance	Mark
8(i)	An explanation that includes two of the following points: • (by causing) {immune/ <u>inflammatory</u>} response (1) • (due to) release of histamine (1) • by causing damage to {blood capillary endothelium / blood vessel / blood brain barrier} (1) • leading to {leaking of fluid from the vessels / (brain) bleeding / increased oedema / increased tissue fluid formation} (1)	ACCEPT humoral response ignore inflammation / inflammation response ACCEPT increased permeability of capillary / weaker blood vessels ACCEPT vasodilation / dilating blood vessel ACCEPT increased {blood flow / fluid in region}	(2)

Question number	Answer	Additional guidance	Mark
8(j)	An answer that includes one of the following points: • {prevent / slow} the {development / progression} of the disease / prevent symptoms appearing / treat before symptoms appear (1) • reverse the effects of the plaques on the brain before they cause damage / prevent plaque formation / {prevent / reduce} amyloid build up (1)	Eg. people are being diagnosed earlier due to better detection techniques Eg. because there could be {plaques / amyloid} with no symptoms Eg. patient to lead a more normal life / work / family Eg. cognitive impairment happens before symptoms detected – Eg. {remove / clear} {plaques / amyloid} (earlier in disease) ignore responses such as better safe than sorry	(1)

